Waddell TR A #3113

Drilling Prognosis for Savanna #417

Surface:	**2227' FSL, 2227' FEL**	**AFE #**	**CX3727**
Bottom:	*2227' FSL, 2227' FEL*	**Well ID #**	**674**
	Sec 19 (A1255), Blk B-21	**Permit #**	**868499**
	PSLS	**API #**	**42-103-36781**
LAT/LON (NAD83):	*31.506538, -102.663069*	**Elevation**	**GL 2637', KB 2649' (12' AGL)**
LAT/LON (NAD27):	*31.506402, -102.662643*	**Lease:**	Waddell TR A
County:	**Crane**	**Field:**	Sand Hills (San Angelo, Upper; Clear Fork; Tubb; Wolfcamp)

Geology		**Casing & Cement**	**Wellhead**	**General Notes**
TVD	*Formation*		**GE API Head**	
			11"	**Notify RRC prior to spud & prior to running & cementing all casing strings**
Near Surface	Red Beds			Drill out with 8.6 ppg FW, vis 33-36, pH > 9.0, no WL control
		Lead: 190 sx 12.8ppg Class C Top of Lead @ 0' Lead Excess@ 680%		WL Totco Inc Survey every 500'
		Tail: 160 sx 13.8ppg Class C Top of Tail @ 175' Tail Excess@ 218%		Add 20sx of starch 300' from TD & circulate through steel pits as needed Pump the rest of the high visc pill when on bottom
	Rustler *Anhydrite*	**8-5/8" 24# J-55 STC**	**475' MD**	**Test Casing to 1,000 psi**
		No deeper than 625' per GAU	**7-7/8"**	**RU H_2S package**
	Salado Salts *Salts*			RU mud loggers at start of production hole Drill out with 10.0 ppg Brine
1,525'	~ Yates *Sands*	Lead: 325 sx 11.6ppg Class C Top of Lead @ 0' Lead Excess@ 142%		**Survey every 200' in the Salts 650' - 2600'**
2,280'	~Queen			**Begin 2-man mud logging over shakers after drilling out of surface casing**
2,750'	~ San Andres *Dolomite*			
3,530'	Glorietta *Sands*	**Marker Joint @ 3500' and 5500' MD**		
3,940'	**Upper Clearfork** *Dolomite*			**Contact Engineer before any trips** Keep hole full during trips
4,380'	Tubb *Sands*	Tail: 960 sx 14.4ppg Class H Top of Tail @ 2450' Tail Excess@ 101%		
5,000'	**Wichita Albany**			
5,610'	**Wolfcamp**			Add 30 sx starch @ 300' from TD & circulate through steel pits <20cc WL, vis 30-32
6,000'	**TVD**	**5-1/2" 17# J-55 LTC**	**6000' MD**	

Approvals

Prepared by: _____
Paiten Tinar, Asset Engineer — Date

Reviewed by: _____
Ian Smith, Permian VP — Date

Waddell - West Ranch Technical Data

Mud Properties

Hole Size	Mud Type	Weight (ppg)	Funnel Visc (s/qt)	PV	YP	FL	pH	Pm	MBT	%LGS
11"	FW / Gel / Lime	8.3 - 8.7	33-36	2-6	2-6	NC	9.0-10	1	-	< 6
11" (300' F/TD)	FW / Gel / Lime / Starch	8.3 - 8.7	33-36	2-6	2-6	<10 cc	9.0-10	1	-	< 6
7-7/8"	Brine / Salt Gel / Lime	10.0-10.2	28-32	1-3	1-3	NC	10	-	-	< 2
7-7/8" (300' F/TD)	Brine / Salt Gel / Lime / Starch	10.0-10.2	30-32		1-3	<20 cc	10	-	-	< 2

Bit & BHA Information

Hole Size	Bit Type	Vendor	Model	Jets	TFA (in^2)	ROP Target	BHA
11"	PDC	Caprock	CR519-SR-8	8x16s	1.38	200 ft/hr	Bit > Bit sub & float > 3 x 8" DCs > XO > 20 x 6.25" DCs > 4-1/2" 16.6# X-95 EU DP
7-7/8"	PDC	Caprock	CR616+SR+K	6x18s	1.491	100ft/hr	Bit > Motor w/float > Monel w/ MWD survey tool > 7-7/8" Stabilizer >1 x 6.5" DC > 7-7/8" Stabilizer > 9 x 6.25" DCs > 10 x 4-1/2" HWDP > 4-1/2" 16.6# G-105 EU DP

**Check with Engr to confirm bit vendor ** All parameters are recommended not hard limits

Motor Specs

Size	Stages	Vendor	Speed, Rev/Gal	Flow Range	Speed Range	Max Diff Press	Max Torque	Bit to Bend	
6-1/2"	7:8 3.0	TALLY	0.155	300-600 gpm	46-93 RPM	1010	16200 ft-lbs	1.5 deg	

Casing Data

Interval	Size (in)	Grade	Weight (ppf)	Thread	ID (in)	Drift (in)	Cap (bbl / ft)	Collapse (psi)		Burst (psi)		Tension (1000 lb)		Make-up (ft-lb)		
								Rated	80%	Rated	80%	Rated	80%	Min	Opt	Max
Surface	8-5/8	J-55	24	STC	8.097	7.972	0.0636	1370	1096	2950	2360	244	195	1830	2440	3050
Prod	5-1/2	J-55	17	LTC	4.892	4.767	0.0232	4910	3928	5320	4256	247	197	1850	2470	3090

Centralizer Program

Casing String	Interval	Spacing	Est Joint Count	Est Turbolizers	Est Centralizers
8-5/8"	1st and 2nd joints	1 per joint	2	0	2
8-5/8"	Thereafter to surface	1 every 4th collar	10	0	3
8-5/8" - Total			**12**	**0**	**5**
5-1/2"	1st and 2nd joints	1 per joint	2	0	2
5-1/2"	Thereafter to Top of Tail CMT	1 every 3rd collar	82	0	28
5-1/2"	Top of tail CMT - surface	1 every 6th collar	58	0	10
5-1/2" - Total			**142**	**0**	**40**

Cement Program

Casing	Slurry	Volume (sx)	Cement	Weight (ppg)	Yield (ft^3/sk)	Water (gal/sk)	Volume (bbls)
8-5/8"	Lead	190	Class C	12.8	1.82	9.77	62
8-5/8"	Tail	160	Class C	13.8	1.60	7.88	46
5-1/2"	Lead	325	50/50 Class C	11.6	2.82	16.4	163
5-1/2"	Tail	960	50/50 Class H	14.4	1.29	5.89	221

Wellhead Information

Section	Bottom	Top	Slips
A - Section (CSGHD)	8-5/8" SOW	11" 3M Flange	5-1/2"
B - Section (TBGHD)	11" 3M Flange	7-1/16" 5M Flange	N/A

Notable changes

Waddell TR A #3113 Interval Specific Parameter Map

11" Surface (CR519-SR-8)

Start	End	ROP Target	WOB Min	WOB Max	WOB Target	Rotary Min	Rotary Max	Rotary Target	GPM Min	GPM Max	GPM Target	Motor Diff	Total RPM	Interval Notes
0-500'	Surf	Max	10	30	**20**	85	120	**110**	400	500	**500**	NA	110	Reach targets as soon as 8" collars clear conductor.
500-475'	Red Beds	Max	25	40	**35**	105	150	**130**	400	500	**500**	NA	130	

7-7/8" Production (CR616+SR+K)

Depths	Zone	ROP Range	WOB Min	WOB Max	WOB Target	Rotary Min	Rotary Max	Rotary Target	GPM Min	GPM Max	GPM Target	Motor Diff Target	Total RPM (.16rev/gal)	Torque, ft-lbs	Interval Notes
475-1525'	Rustler-Yates	**200-400**	20	25	20	40	60	**60**	450	550	**550**	**400+**	145	4-6k	Salado Salt formation, ensure to drill out with 10 ppg Brine.
1475-1575'	Yates Deviation Zone	**200-300**	25	30	25	55	60	**60**	450	550	**500**	**400+**	140	4-6k	Deviations seen in the Salts especially around 1200'-1500'. Once past deviation zone, continue to watch for deviation issues.
1575'-2700'	Yates-San Andres	**150-300**	25	30	25	55	60	**60**	450	550	**500**	**400+**	140	4-6k	
2700'-2800'	Transition into San Andres	**150-200**	30	30	30	40	40	**40**	450	550	**500**	500+	120	4-6k	Transition 50' before to 50' past San Andres.
2800'-3480'	San Andres - Glorieta	**200-300**	30	35	35	55	60	**60**	450	550	**500**	**500+**	140	5-7k	Once San Andres is reached by verification of cuttings, continue to regular drilling.
3480-3580'	Transition into Glorieta	**150-200**	30	30	30	40	40	**40**	450	550	**500**	500+	120	6-7k	Transition 50' before to 50' past Glorieta.
3580-3890'	Glorieta - Upper Clearfork	**200-300**	30	35	35	55	60	**60**	450	550	**500**	**400+**	140	6-7k	
3890-3990'	Transition into Upper Clearfork	**100-150**	30	30	30	40	40	**40**	450	550	**500**	**500+**	120	6-8k	Transition 50' before to 50' past Upper Clearfork.
3990-4330'	Upper Clearfork - Tubb	**150-200**	30	35	35	55	60	**60**	450	550	**500**	**500+**	140	6-8k	
4330-4430'	Transition into Tubb - Lower Clearfork	**100-200**	30	30	30	40	40	**40**	450	550	**500**	500+	120	6-8k	Transition 50' before to 50' past Tubb.
4430-4950'	Tubb - Lower Clearfork	**100-150**	30	35	35	55	60	**60**	450	550	**500**	**400+**	140	8-9k	High WOB and Low RPMS will keep bit engaged in ratty formation
4950-5050'	Transition into Wichita Albany	**100-150**	30	30	30	40	40	**40**	450	550	**500**	**500+**	120	6-8k	Severe losses are possible
4470-5510'	Wichita-Albany	**100-150**	30	35	35	55	60	**60**	450	550	**500**	**500+**	140	6-8k	
5510-5710'	Transition into Wolfcamp	**100-150**	30	30	30	40	40	**40**	450	550	**500**	500+	120	6-8k	Transition 100' before to 100' past Wolfcamp.
5710-6000'	Wolfcamp - TD	**100-150**	30	35	35	55	60	**60**	450	550	**500**	**500+**	140	6-8k	Deviation is possible during the last 200-300'. Discuss with Engineering. Sliding Operations have resulted in sticking the BHA.

Waddell TR A #3113 Contact List

Savanna #417

Well Site Contacts		
Personnel	**Contact**	**Phone**
Toolpusher		
Toolpusher		
Rig Supervisor (Day)	Clint Valentine	337-380-8846
Rig Supervisor (Day)	Bryan Stringer	337-257-3279

Vendor Contacts				
Services	**Company/Person**	**Location**	**Phone**	**Line**
Hospital/Emergency	Ward Memorial Hospital	406 S. Gary Ave Monahans, X 79756	911 432-943-6703	Emergency Office
Sheriff	Ward County Sherriff's Office	300 East 4th Monahans, TX 79756	432-943-6703	Office
Mud Loggers	Geo-site	Midland, TX	325-655-4356	Cell
EDR (Electronic Drilling Recorder)	Pason	Midland, TX	806-281-8370	Cell
8-5/8 Surface Casing (minimum 2 day notice)	Tubular Synergy Group	Midland, TX	432-559-4742	Cell
5-1/2" Production Casing (minimum 2 day notice)	Tubular Synergy Group	Midland, TX	432-559-4743	Cell
Bits	Cap Rock	Odessa, TX	806-201-1706	Cell
Survey Tools	Trueshot	Midland, TX	972-505-0433	Cell
Motor	Tally	Midland, TX	713-443-4669	Cell
Brine Water	Best of Brine	Monahans, Tx	432-631-2317	Cell
Rig Mover	**Bid out: Integrity Services**	Midland, TX	432-556-9563	Office
Casing Crews / LD	**Bid out:**			
Casing Inspection Services	**Bid out:**			
Cementing Services	Pro-Petro	Midland, TX	432-425-4881	Cell
Centralizers	Innovex	Midland, TX	432-250-9652	Cell
Welder	Double A Welding	Monahans, Tx	432-208-5244	Cell
DC Inspection	**Bid Out**			
Diesel	Pilot Thomas	Midland, TX	432-813-6643	Cell
Dirt Contractor/ Pit lining	Ian Barry - Blackbeard Contact	Midland, TX	713-248-4672	Cell
DP Corrosion	Savanna/Corrosion LTD	Midland		
Drilling Mud	Horizon Mud	Midland, TX	432-425-0684	Cell
Float Equipment	Innovex	Midland, TX	432-250-9652	Cell
H_2S Safety Equipment	American Safety Services	Midland, TX	432-664-0451	Cell
Pit Lining / Water Line	Ian Barry - Blackbeard Contact	Midland, TX	713-248-4672	Cell
Rotating Head/Stabilizers	Cutter Drilling Systems	Midland, TX	432-631-4605	Cell
Conductor/ Anchors	Ian Barry - Blackbeard Contact	Midland, TX	713-248-4672	Cell
Texas RRC	District 8 & 8A	Midland, TX	432-684-5581 432-684-6005	Office
Trailer House/ Porta-potties-Trash	Oil Patch	Midland, TX	432-582-4058	Cell
Hot-shot / Trucking	Lonestar Trucking	Midland, TX	432-208-0770	Cell
Hot-shot / Trucking	Driven 2 Deliver Trucking	Midland, TX	432-438-5496	Cell
Fishing Tool	WFR	Midland, TX	575-631-4075	Cell
Wellhead Equipment	Basin Pump Down	Midland, TX	432-250-2003	Cell

Office Contacts				
BBO Personnel	**Contact**	**Cell #**	**Office #**	**Email**
Asset Engineer	Paiten Tinar	210-215-7376		ptinar@blackbeardoperating.com
Asset Engineer/Drilling Advisor	Gus Yanez	903-245-3079		gyanez@blackbeardoperating.com
Asset Engineer	Chase Bosley	210-262-3020		cbosley@blackbeardoperating.com
Asset Engineer	Kevin Flahive-Foro	303-884-2407		kforo@blackbeardoperating.com
Permian VP Operations	Ian Smith	432-528-9628		Ismith@blackbeardoperating.com
Field Foreman	Kenny Bradford	432-813-7985		kbradford@blackbeardoperating.com

General Drilling Procedure

Waddell - West Ranch Drilling Procedure

This program contains the guiding procedures to be followed in the development of this property. It is meant only as a guide under normal-predicted conditions during the drilling operation of this well. As actual conditions differ from predicted, the Drilling Foreman on location, representing Blackbeard Operating, LLC will take appropriate actions needed to continue the drilling operation in the safest and most economical method possible. All changes will be reported to the drilling/operations engineer as soon as drilling operations allow.
The comments, procedures, methods and equipment designs within this document are proprietary to Blackbeard Operating, LLC. Any use of said materials and procedures without authorization from Blackbeard Operating, LLC is prohibited. Please read COVID-19 Protocol below.

Surface Hole

Prep location

1 Preview location prior to rig mobilization to confirm that dimensions meet operational needs, the conductor is set, and the rat hole and mouse hole are drilled.

2 **Read all permits and regulatory paperwork carefully to ensure all regulatory requirements are satisfied.**

3 **Notify RRC 24 hours prior to spudding, running casing, and cementing.** Note the date, time, person contacted, and job # on the daily report.

4 Tubular requirement notifications should be made to pipe vendor in a timely fashion (minimum 2 days notice). Clean, drift, visually inspect, and tally all casing on location.

Drill & survey surface hole

5 a. PU 11" surface bit and BHA per the Technical Data sheet (unless otherwise instructed by the Midland Office). Drill out with fresh water. Must run floats in BHA per TRRC. **Take inclination survey within 500' from surface (all surveys thereafter 500'-1000').** If inclination exceeds 1.5°, back off weight and bring back towards vertical. (Increase the number of surveys to 1 per 300' if inclination exceeds 2.5° resume 1 per 500' when inclination is under 1.0°.)

c. Maintain mud properties per Technical Data Sheet. Circulate inner reserve pit and allow native mud to build funnel vis to ~32-34s. Only go into steel pits if necessary to control losses.

d If Necessary - Mix 2 sx paper per 100' drilled. From slug pit, pump ~10 bbls of high visc sweep every 100'-200' drilled. Circulate and pump paper sweeps, as necessary, before connections to ensure hole is clean and avoid packoffs.

6 a. Drill to match casing strap. Do not set casing more than 200' below the protected depth specified in the GAU Water Board Letter unless an approved Application for Alternate Surface Casing Program is attached.

b. Circulate through steel pits and add 20 sx starch to <10cc water loss 300' before TD.

7 a. At TD circulate hole clean.

b. TOH at an even and steady pace. The Redbeds in this field can swell and become very tight. If this happens, pull very slow and steady till pipe becomes free again. Use surface jars if necessary. **Do not pull more than DC weight.**

Run surface casing

8 Run 1 joint between the guide shoe and the float collar. Thread lock all connections from the guide shoe to the top of the float collar. Use Bestolife 2000 thread compound (or equivalent) on the remaining connections.

9 Install centralizers as specified on Technical Data Sheet.

10 Ensure a casing swage is on the floor to wash down casing as necessary. Keep casing as full as possible to help maintain weight. If you cannot wash casing down, call Engineering to discuss

11 Tag bottom and pick up two feet, circulate one to two casing volumes at max **loss-free** rate.

Cement surface casing

12 Ensure cementing company has analyzed water samples for cementing.

13 Pump 20 bbl FW ahead of cement. Pump cement slurry as defined in the Technical Data sheet. Catch wet and dry samples throughout job, sending dry samples to Midland if a problem arises. Plan to circulate cement to surface.

14 Displace with FW. Bump the plug with 5**00 psi** over final displacement pressure and hold for +/- 5 minutes. DO NOT OVERDISPLACE. Bleed off pressure and check float equipment. Send a copy or ensure that cementer sends a copy of pressure chart from job to Midland Office for well file.

15 **If cement does not circulate to surface, notify the RRC; prepare for a temp survey and 1" top out job.**

NU WH & BOP

17 Install 8-5/8" SOW x 11" 3M casing head, weld inside & outside, test void to 1,000 psi for a minimum of 5 min.

18 Install 11" 5M BOP stack consisting of a drilling spool with choke and kill lines, double rams with pipe rams on top and blind rams on bottom, and an annular preventer with room for a rotating head on top. Should room not be available for the drilling spool, tie choke and kill lines to side outlets below blind rams.

19 Test blind rams, manifold, and valves to **Low 250psi & High 1,500psi** prior to drilling out. Test manifold first & use a test plug if WOC time has not been reached (6hrs for Tail 500psi UCS). Keep a copy of test chart in well file for reference throughout the well. Inspect accumulator unit to ensure that precharge pressures and oil levels are within API specifications. Record tests and inspections on the IADC tour report. Line up for hard shut-in (Choke and first valve downstream of choke in closed position).

20 WOC for a total of 6 hrs before testing casing (500 psi for tail UCS). Prior to drilling out, pressure test the casing to **1000 psig / 30 min per TRRC. If more than 10% pressure is lost during test, call Midland Office for plan forward. Keep a copy of test chart in well file for reference throughout the well.**

Production Hole

Drill production hole NOTE: Loss Remediation procedure updated at the end of the procedure.

1 **R/U a rotating head and H_2S risk mitigation package.** .

2 **a.** No FIT test required IF all of the following requirements are met: 1) CMT to surface 2) Wet samples set up 3) Hard CMT in shoe 4) Compressive strength wait time is met. Contact office to discuss as needed.

b. Test annular and pipe rams to 250psi low and high 1500psi (positive test). Record tests and inspections on the IADC tour report and BBO drilling report. Drill out with brine water.

c. Drill out with a 7-7/8" bit and BHA per the Technical Data sheet (unless otherwise instructed by the Midland Office). Drill out with brine water (take sample of brine water from trucks and have mud engineer test chloride level and list on report). Must run floats in BHA per TRRC. Run inclination only survey tool in BHA & take survey 200' out from casing.

d. When drilling with a motor, differential pressure should be (300-500 psi).

e. Operate pipe rams daily and blind rams on trips.

3 Follow parameter sheet while drilling through the Salado (historical deviation hits around 2,400'). After the Salado, run surveys not to exceed 500' intervals for the entire well. If inclination exceeds 1.5° below surface casing, back off weight and bring back towards vertical.

4 Losses are not expected, but use fibrous materials as needed to control seepage and/or lost circulation. Pump viscous sweeps as needed for hole cleaning. Only add MF-55 to flowline (trickle); DO NOT add in DP.

5 Watch for differential sticking at ALL TIMES. Keep rotary turning or pipe moving as much as possible. On connections pick up off the slips & move the string before bringing the pump up to speed. If you get stuck don't overpull, work the string down and attempt to rotate to get loose

6 **If a flow is encountered note depth and amount of flow on daily report. Oil shows and water flows have been seen in offsets. Can have high H2S concentrations with water flows.**

7 Contact Midland Office if tripping is required and/or if ROP drops significantly for an extended time.

8 Add starch to drilling mud 300' from TD according to Tech Data sheet if hole is sticky.

9 At TD circulate hole clean & take final survey. Pump fluid caliper and compary annular volume to CMT volumes. If CMT volumes appear less than annular volume discuss with engineering adding volume to the lead portion of the cmt.

10 POOH LD DP. Monitor pits to see if well is flowing and kill as necessary.

Run production casing

11 **Notify RRC 24 hours prior to running and cementing casing.**

12 Run 1 joint between the float shoe and the float collar. Thread lock all connections from float shoe to top of float collar. Use Bestolife 2000 thread compound (or equivalent) on the remaining connections.

13 Ensure a casing swage is on the floor to wash down casing as necessary.

14 Install centralizers as specified on Technical Data Sheet.

15 Run marker joints as specified on the Well Sketch.

16 Tag bottom, pick up two feet. Circulate the hole and RU cementing head. Circulate the hole a minimum of one full circulation at max **loss-free** rate.

Cement production casing

17 Ensure cementing company has analyzed water samples for cementing.

18 Pump 20 bbls FW pre-flush. Contact Midland for any changes. Pump cement slurry as defined in the Technical Data sheet or as advised by engineering due to excessive fluid caliper. Catch wet and dry samples throughout job, sending dry samples to Midland if a problem arises. Plan to circulate cement to surface.

19 Displace with fresh water. Note on daily report how much cement circulates. We are **not** required to circulate cement on this production string. On wells with no CMT to surface, report 200' TOC per Prog.

20 Bump plug with 500 psi over final displacement pressure. DO NOT OVERDISPLACE. Release pressure and ensure float equipment is holding. Send a copy or ensure that cementer sends a copy of pressure chart from each stage to Midland Office for the well file.

21 If any flow is observed after cement job, wash out BOPs and close in on well for 6 hours minimum.

ND BOP & NU WH, RDMO

22 ND BOP, set slips, NU and test 11" 3M x 7-1/16" 5M tbghd to 3,000 psi for 5 min.

23 Jet and clean the pits, release rig.

Loss Remediation Procedure

1 Upon drilling into a loss zone evaluate the severity of the losses. If offset information indicates there will be losses have a LCM pill premixed before drilling into lose zone. Drill 100ft into loss zone and attempt to heal with the following sweep.

40bbl LCM Sweep:
10Sx of Cotton seed hulls (Coarse)
5Sx of Cedar Fiber (Coarse)
10 Sacks of Salt Gel
5 Sacks of Starch

2 Monitor returns post LCM sweep. If partial returns are seen proceed to work with mud engineer to come up with a game plan to sweep the hole as we drill to try and heal our losses. If we still have full losses and we know of severe losses in offset well than we will plan to cement out losses.

3 Dry drill an additional 300-500ft before coming out of hole to cmt. While coming out of the hole stop 100ft above the loss zone and pump a heavy LCM sweep as follows.

70 bbls LCM Sweep:
20 Sx of cotton seed hulls (coarse)
10 Sacks of Cedar fiber (coarse)
20 sacks of salt gel
5 sacks of starch

4 POOH and LD BHA and Drill collars.

5 RBIH w/ open ended drill pipe. Before exiting surface casing shoe (conductor shoe if cmting surface hole) plan on attempting to establish returns. Continue to RIH w drill pipe and stop ~100ft above loss zone. Rig up cementers and plan on pumping CMT

a Discuss CMT program with engineering. Previous success has been made with a full truck load of CMT ~280 sx of a Class C CMT at 14.2ppg. Discuss with cementers before pumping.

b Establish an injection rate before pumping cement.

c. Mix and pump cmt paying close attention to annular pressure and lift pressure on pump trucks. Displace Cmt into formation.

- If you notice lift pressure or annular pressure while cementing. Cut Cmt and go to flush. Flush 2 annular volumes of FW and prepare to POOH to previous CSG shoe. IF deemed necessary mix and pump another annular volume of FW at shoe before pulling DP out of hole.
- If lift pressure remains constant than displace cmt and flush with 2 annular volumes of FW. POOH W/ DP.

6 Let CMT sit for 8 hours before RIH w/ bit to drill out cmt.

7 After 8 hours tag top of plug and note on report. If top of CMT plug feels very “soft” than considering shutting in an additional 4 hours to allow CMT to harden.

COVID-19 Preventative Measures

1. Everyone entering a Blackbeard drilling location must check in with either the company man or the tool pusher.
   - **a** The Savanna tool-pusher will be accountable for all the Savanna employees.
   - **b** The Company man will be accountable for everyone else.
2. Everyone entering location must be wearing a mask prior to checking in with the company man or tool pusher.
3. Everyone must sanitize their hands prior to signing in.
4. Company man or tool pusher will be required to take the temperature of the person signing in.
   - **a** In the event they test over 100 Degrees F they will either be sent home or allowed to try again after cooling in their own personal space (most likely their vehicle) .
   - **b** If they are below the 100 Degree F threshold than they can sign in and go about their business.
5. Company man and tool pusher will sanitize their office area/sign in area throughout the day.
6. When/if possible, work 6ft apart from one another.
7. A mask will be required when a person must work at a distance of 6ft or less with any other individual.
   - **a** Most of the people working on location can work 6ft apart from each other.
   - **b** Designed for events where multiple people will be working in close proximity.
   - **c** Will be enforced on pre-spud or JSA meetings.

BLACKBEARD OPERATING

Geologic Prognosis - Vertical Well

CX3727

WELL NAME:	Waddell TR A #3113	**FIELD OR PROSPECT:**	Sand Hills (San Angelo, Upper; Clear Fork; Tubb; Wolfcamp)
SURFACE LOCATION:	2227' FSL, 2227' FEL, Sec 19 (A1255), Blk B-21, Crane, TX		
API Number:	42-103-36781	**RRC Permit Number:** 868499	**Drilling Rig:** Savanna #417
Driving Directions:	***SEE ATTACHED MAP FOR DIRECTIONS***		
LAT/LON (NAD83):	31.506538, -102.663069	**KB:**	2,649'
LAT/LON (NAD27):	31.506402, -102.662643	**GROUND ELEVATION:**	2,637'
PRIMARY OBJECTIVES:	Wolfcamp	**PROPOSED TOTAL DEPTH:**	6,000'
SECONDARY OBJECTIVES:	Glorieta/Clearfork/Tubb		
KNOWN DRILLING HAZARDS:	Rustler Anhydrite, possible lost circulation. Yates, possible water flow. See depths below.		
OPEN-HOLE LOGGING PROGRAM:	None		
RECOMMENDED CORING AND TESTING PROGRAM:	None		
WELLSITE SUPERVISION:	Clint Valentine Cell - 337-380-8846 or Bryan Stringer Cell - 337-257-3279		
MUD LOGGER:	Geo-site Cell - 325-655-4356 *Start 2 man mud logging after drilling out of surface casing		
GEOLOGIST:	Rene Wiksveen 432-230-7629, rwiksveen@blackbeardoperating.com		

ESTIMATED GEOLOGIC TOPS

	SUBJECT WELL		PRIMARY CORRELATION WELLS (NAME, LOC.):				
	EST. KB:	12'	Waddell TR A 1901 (1031' SW)		Waddell TR A 3126 (2587' E)		
	ELEV_KB:	2,649'	**ELEV_KB:**	2,629'	**ELEV_KB:**	2,646'	
FORMATION	DEPTH	SS	DEPTH	SS	DEPTH	SS	COMMENTS
GAU	425'	2,224'	425'	2,204'	425'	2,221'	*Water board Letter*
SWR 13 Excptn Letter		2,649'	'	2,629'	'	2,646'	*Possible lost circulation*
Yates	1,525'	1,124'	1,499'	1,130'	1,549'	1,097'	
Queen	2,280'	369'	2,257'	372'	2,328'	318'	*Set 8-5/8" surface casing*
San Andres	2,750'	-101'	2,712'	-83'	2,788'	-142'	*Possible water flow*
** Glorieta	3,530'	-881'	3,499'	-870'	3,613'	-967'	
** Upper Clearfork	3,940'	-1,291'	3,909'	-1,280'	4,003'	-1,357'	
Tubb	4,380'	-1,731'	4,352'	-1,723'	4,354'	-1,708'	
** Lower Clearfork	4,420'	-1,771'	4,394'	-1,765'	4,406'	-1,760'	
Wichita-Albany	5,000'	-2,351'	4,976'	-2,347'	4,904'	-2,258'	
* Wolfcamp	5,610'	-2,961'	5,589'	-2,960'	5,626'	-2,980'	
Wolfcamp Base	5,820'	-3,171'	5,801'	-3,172'	5,858'	-3,212'	
Waddell	-	-	-	-	-	-	
Connell	-	-	-	-	5,859'	-3,213'	
Joins	-	-	-	-	5,860'	-3,214'	
Ellenburger	5,975'	-3,326'	5,954'	-3,325'	5,931'	-3,285'	
Total Depth	**6,000'**	**-3,351'**	**6,000'**	**-3,371'**	**6,150'**	**-3,504'**	

* *Primary Objective*
** *Secondary Objective*

Production Casing Cement - Volumes to be adjusted according to fluid caliper logs, along with actual casing depths. Plan top of tail cement to come 300ft above the top of the San Andres Formation.

MUDLOGGING REQUIREMENTS

Interval: BSC to TD, sample every 30', keep dry samples
Gas Readings: ***Begin 2-man mud logging over shakers after drilling out of surface casing***
If gas readings are not consistent with offset wells, please perform necessary equipment testing and calibration
Reporting: Send one Daily PDF Report covering the last 24 hours with your morning email update
Send email update at least 4 times daily (5 & 11 AM/PM). Include 1"/5" PDFs, 5" TIFF, LAS in every update.
Notify Geologist immediately if there are any mechanical or technical issues ***call/text/email***
Annotation Please annotate presence/lack of FLU and CUT on the mudlog; including oil in cuttings/pit/etc
Please annotate any mechanical issues affecting the gas/cuttings data collection on the mudlog
Images: Send images of cuttings (hourly chart), cuttings trays, and cut per well
Datum: NAD27
Final Logs: Send final digital log email to distribution list
Send 2x Color prints to Paiten Tinar, 200 N. Loraine, Ste 300, Midland, TX 79701
Cuttings: Send cuttings to The Midland Subsurface Library

Railroad Commission of Texas

PERMIT TO DRILL, RE-COMPLETE, OR RE-ENTER ON REGULAR OR ADMINISTRATIVE EXCEPTION LOCATION

CONDITIONS AND INSTRUCTIONS

Permit Invalidation. It is the operator's responsibility to make sure that the permitted location complies with Commission density and spacing rules in effect on the spud date. The permit becomes invalid automatically if, because of a field rule change or the drilling of another well, the stated location is not in compliance with Commission field rules on the spud date. If this occurs, application for an exception to Statewide Rules 37 and 38 must be made and a special permit granted prior to spudding. Failure to do so may result in an allowable not being assigned and/or enforcement procedures being initiated.

Notice Requirements. Per H.B 630, signed May 8, 2007, the operator is required to provide notice to the surface owner no later than the 15th business day after the Commission issues a permit to drill. Please refer to subchapter Q Sec. 91.751-91.755 of the Texas Natural Resources Code for applicability.

Permit expiration. This permit expires two (2) years from the date of issuance shown on the original permit. The permit period will not be extended.

Drilling Permit Number. The drilling permit number shown on the permit MUST be given as a reference with any notification to the district (see below), correspondence, or application concerning this permit.

Rule 37 Exception Permits. This Statewide Rule 37 exception permit is granted under either provision Rule 37 (h)(2)(A) or 37(h)(2)(B). Be advised that a permit granted under Rule 37(h)(2)(A), notice of application, is subject to the General Rules of Practice and Procedures and if a protest is received under Section 1.3, "Filing of Documents," and/or Section 1.4, "Computation of Time," the permit may be deemed invalid.

Before Drilling

Fresh Water Sand Protection. The operator must set and cement sufficient surface casing to protect all usable-quality water, as defined by the Railroad Commission of Texas (RRC) Groundwater Advisory Unit (GWAU). Before drilling a well, the operator must obtain a letter from the Railroad Commission of Texas stating the depth to which water needs protection, Write: Railroad Commission of Texas, Groundwater Advisory Unit (GWAU), P.O. Box 12967, Austin, TX 78711-3087. File a copy of the letter with the appropriate district office.

Accessing the Well Site. If an OPERATOR, well equipment TRANSPORTER or WELL service provider must access the well site from a roadway on the state highway system (Interstate, U.S. Highway, State Highway, Farm-to-Market Road, Ranch-to-Market Road, etc.), an access permit is required from TxDOT. Permit applications are submitted to the respective TxDOT Area Office serving the county where the well is located.

Water Transport to Well Site. If an operator intends to transport water to the well site through a temporary pipeline laid above ground on the state's right-of-way, an additional TxDOT permit is required. Permit applications are submitted to the respective TxDOT Area Office serving the county where the well is located.

*NOTIFICATION

The operator is **REQUIRED** to notify the district office when setting surface casing, intermediate casing, and production casing, or when plugging a dry hole. The district office MUST also be notified if the operator intends to re-enter a plugged well or re-complete a well into a different regulatory field. Time requirements are given below. The drilling permit number MUST be given with such notifications.

During Drilling

Permit at Drilling Site : A copy of the Form W-1 Drilling Permit Application, the location plat, a copy of Statewide Rule 13 alternate surface casing setting depth approval from the district office, if applicable, and this drilling permit must be kept at the permitted well site throughout drilling operations.

***Notification of Setting Casing :** The operator MUST call in notification to the appropriate district office (phone number shown the on permit) a minimum of eight (8) hours prior to the setting of surface casing, intermediate casing, AND production casing. The individual giving notification MUST be able to advise the district office of the drilling permit number.

***Notification of Re-completion/Re-entry :** The operator MUST call in notification to the appropriate district office (phone number shown on permit) a minimum of eight (8) hours prior to the initiation of drilling or re-completion operations. The individual giving notification MUST be able to advise the district office of the drilling permit number.

Completion and Plugging Reports

Hydraulic Fracture Stimulation using Diesel Fuel: Most operators in Texas do not use diesel fuel in hydraulic fracturing fluids. Section 322 of the Energy Policy Act of 2005 amended the Underground Injection Control (UIC) portion of the federal Safe Drinking Water Act (42 USC 300h(d)) to define "underground Injection" to *EXCLUDE* " ...the underground injection of fluids or propping agents *(other than diesel fluels)* pursuant to hydraulic fracturing operations related to oil, gas, or geothermal production activities." (italic and underlining added.) Therefore, hydraulic fracturing may be subject to regulation under the federal UIC regulations if diesel fuel is injected or used as a propping agent. EPA defined "diesel fuel" using the following five (5) Chemical Abstract Service numbers: 68334-30-5 Primary Name: Fuels, diesel; 68476-34-6 Primary Name: Fuels, diesel, No. 2; 68476-30-2 Primary Name: Fuel oil No. 2; 68476-31-3 Primary Nmae: Fuel oil, No. 4; and 8008-20-6 Primary Name: Kerosene. As a result, an injection well permit would be required before performing hydraulic fracture stimulation using diesel fuel as defined by EPA on any well in Texas. Hydraulic fracture stimulation using diesel fuel as defined by EPA on a well in Texas without an injection well permit could result in enforcement action.

Producing Well : Statewide Rule 16 states that the operator of a well shall file with the Commission the appropriate completion report within thirty (30) days after completion of the well or within ninety (90) days after the date on which the drilling operation is completed, whichever is earlier. Completion of the well in a field authorized by this permit voids the permit for all other fields included in the permit unless the operator indicates on the initial completion report that the well is to be a dual or multiple completion and promptly submits an application for multiple completion. All zones are required to be completed before the expiration date on the existing permit. Statewide Rule 40(d) requires that upon successful completion of a well in the same reservoir as any other well previously assigned the same acreage, proration plats and P-15s (if required) must be submitted with no double assignment of acreage.

Dry or Noncommercial Hole : Statewide Rule 14(b)(2) prohibits suspension of operations on each dry or non-commercial well without plugging unless the hole is cased and the casing is cemented in compliance with Commission rules. If properly cased, Statewide Rule 14(b)(2) requires that plugging operations must begin within a period of one (1) year after drilling or operations have ceased. Plugging operations must proceed with due diligence until completed. An extension to the one-year plugging requirement may be granted under the provisions stated in Statewide Rule 14(b)(2).

Intention to Plug : The operator must file a Form W-3A (Notice of Intention to Plug and Abandon) with the district office at least five (5) days prior to beginning plugging operations. If, however, a drilling rig is already at work on location and ready to begin plugging operations, the district director or the director's delegate may waive this requirement upon request, and verbally approve the proposed plugging procedures.

***Notification of Plugging a Dry Hole :** The operator MUST call in notification to the appropriate district office (phone number shown on permit) a minimum of four (4) hours prior to beginning plugging operations. The individual giving the notification MUST be able to advise the district office of the drilling permit number and all water protection depths for that location as stated in the Texas Commission on Environmental Quality letter.

DIRECT INQUIRIES TO: DRILLING PERMIT SECTION, OIL AND GAS DIVISION

PHONE
(512) 463-6751

MAIL:
PO Box 12967
Austin, Texas, 78711-2967

RAILROAD COMMISSION OF TEXAS
OIL & GAS DIVISION

PERMIT TO DRILL, DEEPEN, PLUG BACK, OR RE-ENTER ON A REGULAR OR ADMINISTRATIVE EXCEPTION LOCATION

PERMIT NUMBER	DATE PERMIT ISSUED OR AMENDED	DISTRICT
868499	03/24/2021	08
API NUMBER	**FORM W-1 RECEIVED**	**COUNTY**
42-103-36781	03/19/2021	CRANE
TYPE OF OPERATION	**WELLBORE PROFILE(S)**	**ACRES**
New Drill	Vertical	23065.0

OPERATOR 073056

BLACKBEARD OPERATING, LLC

1751 RIVER RUN
SUITE 405
FORT WORTH, TX 76107-0000

NOTICE
This permit and any allowable assigned may be revoked if payment for fee(s) submitted to the Commission is not honored.
District Office Telephone No:

(432) 684-5581

LEASE NAME	WELL NUMBER
WADDELL TR A	3113
LOCATION	**TOTAL DEPTH**
20.7 miles NW direction from Crane	9000

Section, Block and/or

SECTION 19 BLOCK B21 ABSTRACT 1255

SURVEY PSL / WADDELL, W N & CO

DISTANCE TO SURVEY LINES	DISTANCE TO NEAREST LEASE LINE
2227.0 ft S 2227.0 ft E	null
DISTANCE TO LEASE LINES	**DISTANCE TO NEAREST WELL ON LEASE**
7586.0 ft S 18895.0 ft W	See FIELD(s) Below

FIELD(s) and LIMITATIONS:

* SEE FIELD DISTRICT FOR REPORTING PURPOSES *

FIELD NAME / LEASE NAME	ACRES / NEAREST LEASE	DEPTH	WELL# / NEAREST WELL	DIST
SAND HILLS (CLEAR FORK)	23065.0	9000	3113	08
WADDELL TR A			5201.0	

This is a hydrogen sulfide field. This well shall be drilled in accordance with SWR 36.

No allowable will be assigned until the acreage on this W-1 (23,065) has been subdivided out of the WADDELL, W. N., ETAL lease. The new name will be WADDELL TR A.

SAND HILLS (SAN ANGELO, UPPER)	23065.0	9000	3113	08
WADDELL TR A			1262.0	

This is a hydrogen sulfide field. This well shall be drilled in accordance with SWR 36.

No allowable will be assigned until the acreage on this W-1 (23,065) has been subdivided out of the WADDELL, W. N., ETAL lease. The new name will be WADDELL TR A.

SAND HILLS (TUBB)	23065.0	9000	3113	08
WADDELL TR A			2583.0	

This is a hydrogen sulfide field. This well shall be drilled in accordance with SWR 36.

No allowable will be assigned until the acreage on this W-1 (23,065) has been subdivided out of the WADDELL, W. N., ETAL lease. The new name will be WADDELL TR A.

SAND HILLS (WOLFCAMP)	23065.0	9000	3113	08
WADDELL TR A			2583.0	

This is a hydrogen sulfide field. This well shall be drilled in accordance with SWR 36.

No allowable will be assigned until the acreage on this W-1 (23,065) has been subdivided out of the WADDELL, W. N., ETAL lease. The new name will be WADDELL TR A.

THE FOLLOWING RESTRICTIONS APPLY TO ALL FIELDS

This well shall be completed and produced in compliance with applicable special field or statewide spacing and density rules. If this well is to be used for brine mining, underground storage of liquid hydrocarbons in salt formations, or underground storage of gas in salt formations, a permit for that specific purpose must be obtained from Environmental Services prior to construction, including drilling, of the well in accordance with Statewide Rules 81, 95, and 97.

This well must comply to the new SWR 3.13 requirements concerning the isolation of any potential flow zones and zones with corrosive formation fluids. See approved permit for those formations that have been identified for the county in which you are drilling the well in.

Railroad Commission of Texas
Oil and Gas Division
SWR #13 Formation Data
CRANE (103) COUNTY

Formation	Remarks	Geological Order	Effective Date
RUSTLER		1	02/09/2020
YATES		2	02/09/2020
SEVEN RIVERS		3	02/09/2020
QUEEN		4	02/09/2020
GRAYBURG	active CO2 flood	5	02/09/2020
SAN ANDRES	usable quality water / active CO2 flood	6	02/09/2020
SAN ANGELO		7	02/09/2020
GLORIETA		8	02/09/2020
PERMIAN GENERAL		9	02/09/2020
CLEARFORK		10	02/09/2020
TUBB		11	02/09/2020
WICHITA ALBANY		12	02/09/2020
WOLFCAMP		13	02/09/2020
CISCO		14	02/09/2020
PENNSYLVANIAN		15	02/09/2020
STRAWN		16	02/09/2020
BEND		17	02/09/2020
DEVONIAN	active CO2 flood	18	02/09/2020
FUSSELMAN		19	02/09/2020
MONTOYA		20	02/09/2020
WADDELL		21	02/09/2020
ELLENBURGER		22	02/09/2020
PRECAMBRIAN (UNDIFFERENTIATED)		23	02/09/2020

The above list may not be all inclusive, and may also include formations that do not intersect all wellbores. The listing order of the Formation information reflects the general stratigraphic order and relative geologic age. This is a dynamic list subject to updates and revisions. It is the operator's responsibility to make sure that at the time of spudding the well the most current list is being referenced. Refer to the RRC website at the following address for the most recent information. http://www.rrc.texas.gov/oil-gas/compliance-enforcement/rule-13-geologic-formation-info

GROUNDWATER PROTECTION DETERMINATION

Form GW-2

Groundwater Advisory Unit

Date Issued:	20 August 2018	**GAU Number:**	202536

Attention:	BURLINGTON RES O & G CO- ATTN AMY JOHNSON EC3 10 HOUSTON, TX 77079	**API Number:**	10302267
		County:	CRANE
		Lease Name:	WADDELL, W. N., ETAL
		Lease Number:	
Operator No.:	109336	**Well Number:**	826
		Total Vertical Depth:	3505
		Latitude:	31.502691
		Longitude:	-102.667259
		Datum:	NAD27

Purpose: Recompletion (RC)

Location: Survey-PSL; Abstract-1255; Block-B21; Section-19

To protect usable-quality groundwater at this location, the Groundwater Advisory Unit of the Railroad Commission of Texas recommends:

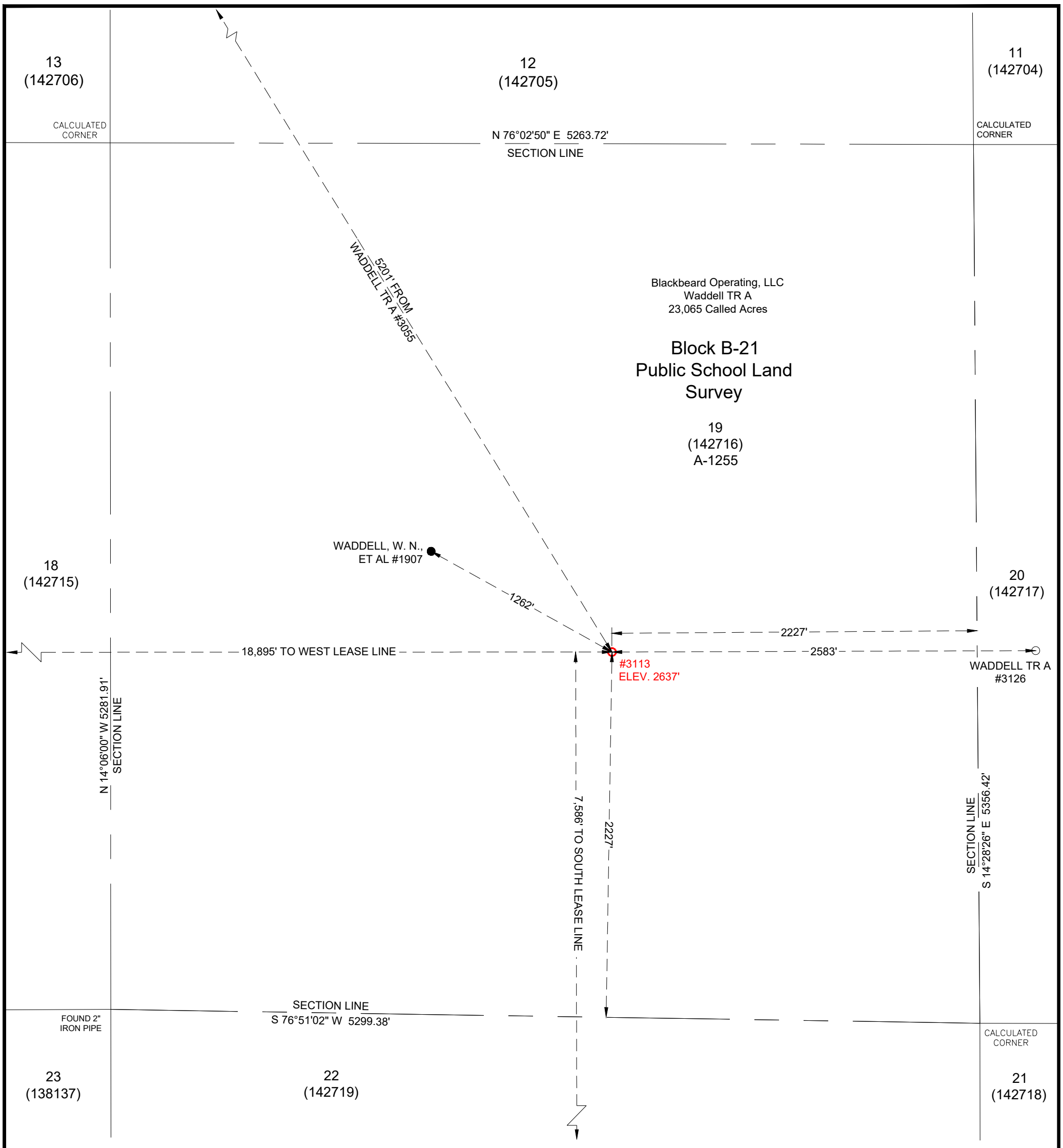
The interval from the land surface to a depth of 425 feet must be protected.

This recommendation is applicable for all wells drilled in this Section 19.

Note: Unless stated otherwise, this recommendation is intended to apply only to the subject well and not for area-wide use. Unless stated otherwise, this recommendation is for normal drilling, production, and plugging operations only.

This determination is based on information provided when the application was submitted on 08/17/2018. If the location information has changed, you must contact the Groundwater Advisory Unit, and submit a new application if necessary. If you have questions, please contact us at 512-463-2741 or gau@rrc.texas.gov.

Groundwater Advisory Unit, Oil and Gas Division



Application for Well No. #3113
20.7 miles Northwest of Crane, TX.

Location Coordinates for Waddell TR A #3113
NAD 27 X = 1274129.86
NAD 27 Y = 676693.79
NAD 27 LAT. = 31°30'23.05"/ 31.506402
NAD 27 LONG. = -102°39'45.52"/ -102.662643
NAD 83 X = 1570596.98
NAD 83 Y = 10519269.40
NAD 83 LAT. = 31°30'23.54"/ 31.506538
NAD 83 LONG. = -102°39'47.05"/ -102.663069

STATE OF TEXAS
REGISTERED
JAMES CURTIS YARGER
5854
PROFESSIONAL
LAND SURVEYOR

EST. 2014
BLACKBEARD
OPERATING
Midland
TX

Driving Directions:

From Crane (int. of HWY 385 & FM 329) to well.
Head West On FM 329 For 15.9 Mi. Turn right onto FM 1053 N For 8.5 Mi. Turn Left onto an existing lease road For 0.5 Mi. Turn Right onto an existing lease road For 0.6 Mi. Destination Will Be On The Right.

Notes:
1) Bearings and coordinates shown hereon are referenced to the Texas Coordinate System of 1983 (Texas Central Zone 4203).
2) Latitude & Longitude are NAD 83 and NAD 27 geographic.
3) This is an exhibit and does not represent a true boundary survey. this survey is based on ownership and easement information provided by Blackbeard Operating, LLC. Surveyor did not abstract subject tract and there may be easement or other encumbrances that affect the subject tract that are not shown hereon.

600' 0' 600'
SCALE: 1" = 600'

I HEREBY STATE THAT THIS SURVEY WAS PERFORMED UNDER MY DIRECT SUPERVISION

JAMES CURTIS YARGER
REGISTERED PROFESSIONAL LAND SURVEYOR
STATE OF TEXAS # 5854

Date Surveyed October 1, 2020

Blackbeard Operating, LLC
Waddell TR A #3113
2227' FSL & 2227' FEL

Section 19 (A-1255), Block B-21,
Public School Land Survey
Crane County, Texas

LEGEND
DENOTES SURVEY BOUNDARY/TRACT LINE
DENOTES LEASE TRACT
DENOTES PROPOSED WELL LOCATION
DENOTES PRODUCING WELL LOCATION
DENOTES PERMITTED WELL LOCATION

TRANSGLOBAL SERVICES LLC
TBPELS FIRM# 10193740 / 19148
2129 S Great Southwest Parkway Suite 313
Grand Prairie, TX 75051
(817) 529-1180 ~ Fax (817) 529-1181

DRAWN BY: MAC	DATE: 12-01-2020
CHECKED BY: JCY	DATE: 03-19-2021
PAGE 1 OF 1	SCALE: 1" = 600'

DWG. NO.
Transglobal Services LLC\Blackbeard Operating - Survey - Documents\20-02062_Waddell TR A #3113

REVISION NO.
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